



A subnet that makes computer-use agents accurate enough to ship.

让计算机使用代理足够精准、可以投产的子网。

We pay open-weight models for one thing only — turning a natural-language instruction and a screen into the **right pixel**. Validators score it. Emissions follow accuracy.

我们只为开源权重模型做一件事付费——把自然语言指令和屏幕变成**正确的像素**。验证器打分，奖励跟随准确率。

SUBNET
DDX · sn candidate

MECHANISM
Defektr-class
(2025 hackathon proposal)

BENCHMARKS
UI-Vision · OSWorld-G · SS-Pro

TIME TO LIVE DEMO
48 hours

TEAM
Drew Mimi
Keith Guo

GROUND TRUTH, ON-CHAIN.

• THESIS 论点

Computer-use agents don't fail at reasoning. They fail at finding the pixel.

计算机使用代理不是输在推理上，而是输在找像素上。

66%

Agent failures traced to grounding errors.

代理失败可追溯到接地错误。

Reasoning is solved-enough. The model picks the wrong coordinate, the

\$139–199B

Agentic AI infrastructure by 2034 (34%+ CAGR).

到 2034 年的代理型 AI 基础设施。

Anthropic, OpenAI, Google all train grounding directly. Frontier hits **86%** on ScreenSpot-Pro — saturating. **UI-Vision** and fine-grained **OSWorld-G**

remain **below 40%** end-to-end. Specialist open models (UI-TARS, OS-

0

Open networks paying miners to win GUI grounding.

为矿工赢得 GUI 接地付费的开放网络。 THE BOTTLENECK MOVED.

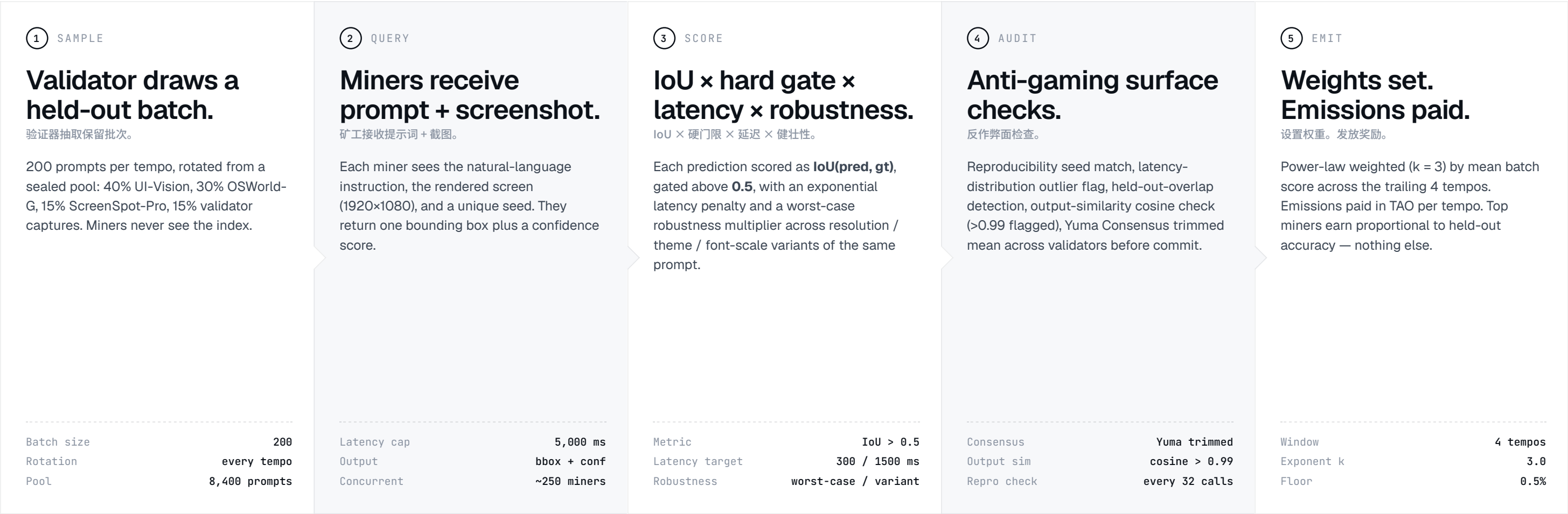
Open-source teams ship in isolation. No coordinated incentive. DDX is the

The validator loop.

验证器循环。

One tempo, end-to-end. Defektr-class mechanism — proven, low-novelty, low surface area to game. Every stage is deterministic and reproducible on standardized reference hardware.

一个节奏，端到端。Defektr 级机制——已验证、创新少、博弈面小。每个阶段都是确定性的，可在标准参考硬件上复现。



A defensible scoring mechanism.

经得起推敲的评分机制。

PER-PREDICTION SCORE

单次预测得分

```
// for miner m, prompt p:
score(m, p)
= IoU(predm,p, gtp)
  × 1[IoU > τ]
  × exp(-λ · latency)
  × robustness(m, p)
```

PER-TEMPO EMISSION WEIGHT

单节奏奖励权重

```
Wm = μ(scorem, B)k / Σj μ(scorej, B)k
// k = 3 · power-law concentration on frontier
```

IoU	Intersection-over-union of predicted & ground-truth bbox. IoU > 0.5 is the stricter variant of click-within-bbox we chose deliberately. 预测与真实边界面框的交集面积比。严于点击命中的变体。
τ	Hard gate at 0.5 . No partial credit for “close”. 硬门限 0.5，“接近”不得分。
λ	Latency soft target 300ms ; decays exponentially, hard ceiling at 1500ms . 300ms 内满分；1500ms 处为零。
robustness	Worst-case score across resolution / theme / font-scaling variants of the same prompt. 同一提示词在多种屏幕变体下的最坏情况得分。
k = 3	Power-law exponent. Concentrates emissions on the frontier while preserving 2nd / 3rd place economics. 幂律指数，集中奖励前沿。

ANTI-GAMING SURFACE

反作弊面

HELD-OUT LEAK

保留集泄露

Ground truth lives only in validator binary. Sealed pool: 40% UI-Vision, 30% OSWorld-G, 15% ScreenSpot-Pro, 15% validator captures. Rotated every tempo.
// leak half-life ~16 tempos · pool size 8,400 prompts

DISTRIBUTION-SHIFT GAMING

分布漂移作弊

15% of every batch is held-out captures from applications outside any public benchmark, refreshed weekly by validators. Pure-benchmark overfitters score poorly here.
// the moat against frontier labs that train on public sets

RESOLUTION / THEME OVERFIT

分辨率 / 主题过拟合

Robustness term scores worst-case across 720p / 1080p / 1440p / 4K, light / dark / high-contrast, 100 / 125 / 150% font scaling. Cannot game one variant without losing the others.
// 36 variant combinations per prompt · cheapest variant sets the score

WEIGHT COPYING

权重复制

Output-level cosine similarity check across all miner predictions. Pairs above 0.99 flag the later-registered miner for pruning.
// detects copy-paste miners without inspecting weights

SYBIL FARMING

女巫农场

Stake-weighted scoring; 0.5% emission floor; new-UID emission delay of 8 tempos to disincentivize churn.
// fresh-UID breakeven: ~10h of held-out accuracy > 0.55

STOCHASTIC ENSEMBLES

随机集成 / 外部 API 代理

Every 32nd call is a seeded re-query. Output mismatches trigger temporary 50% weight penalty until cleared.
// catches ensembles and miners proxying to closed APIs

LATENCY GAMING

延迟作弊

Per-miner latency distribution monitored. >3σ deviation triggers audit and pins miner to median latency pending review.
// no sandbagging until easy prompts arrive

VALIDATOR COLLUSION

验证器串通

Yuma Consensus stake-weighted trimmed mean across validators. Open evaluation code. Standardized reference hardware (RTX 4090, fixed driver) makes outputs reproducible across any validator.
// inherits Bittensor base-layer consensus + reproducibility floor

Four components — IoU, a hard gate, a latency penalty, robustness across screen variants. Nothing learned, nothing tunable per-miner. 200 lines of validator code; weeks of adversarial review.

四个组成部分——IoU、硬门限、延迟惩罚、跨屏幕变体的健壮性。无可学习、无可调参。200 行验证器代码；数周对抗审查。

Three benchmarks. One weighted score.

三个基准。一个加权得分。

Public, instrumented, peer-reviewed. We blend them so no single benchmark dominates — and so miners that overfit to one lose ground on the composite.

UI-V

weight 0.40

UI-Vision

桌面端 · 细粒度接地
Nayak et al. 2025 · 83 desktop applications

Desktop-centric, license-permissive, fine-grained. Element grounding, layout grounding, action prediction. Best end-to-end published score: **36.2%** (Phi-Ground). Where real desktop agents still fail.

APPS

83

TASKS

7,500+

SCORE TYPE

click-in-bbox

CEILING

<40% · headroom

OSW-G

weight 0.30

OSWorld-G

NeurIPS 2025 Spotlight · 精确控件
Xie et al. 2025 · NeurIPS Spotlight

Fine-grained UI manipulation across 32 distinct element types. Tests precise widget interaction that other benchmarks miss. Open-source SOTA: **Qwen3-VL-235B at 68.3%**.

ELEMENT TYPES

32

TASKS

564

SCORE TYPE

click-in-bbox

CEILING

<70% · headroom

SS-PRO

weight 0.30

ScreenSpot-Pro

专业软件 · 4K · 对比基线
Li et al. 2025 · 1,581 high-res instances

Professional software — Photoshop, AutoCAD, Blender, Premiere. 4K resolution, dense UI. Established leaderboard. Frontier: **GPT-5.2 86.3%**. Open-source: **UI-Venus-72B 61.9%**.

APPS

23

RESOLUTION

3840×2160

SCORE TYPE

IoU

OPEN SOTA

62% · less room

UPCOMING v1.1 adds **VisualWebArena fragments** (0.10) and a **private DDX-generated set** (0.10) once 50 validators are live. Weights are governed by stake vote.

VALIDATOR UI

LIVE

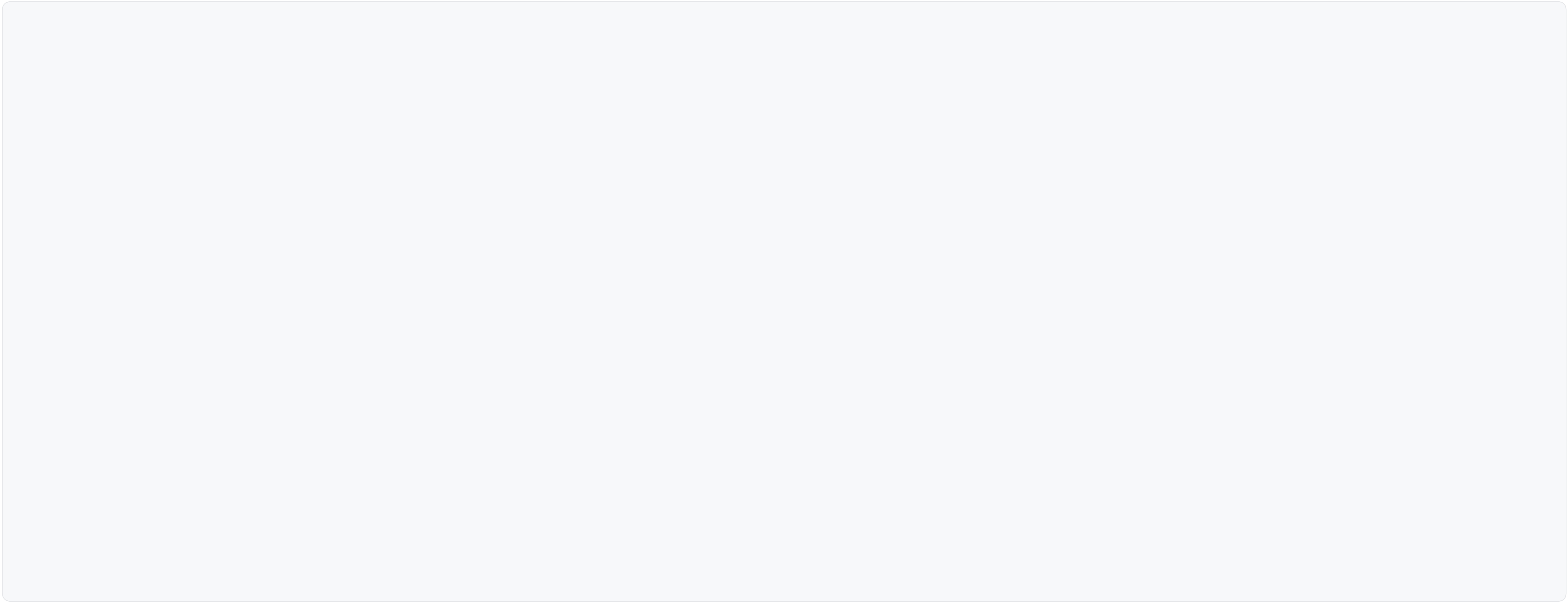
VALI-04.DDX

Anyone can audit the loop, live.

任何人都能实时审计这个循环。

Public read-only validator console. Real-time leaderboard, score history, anti-gaming flags, batch rotation status. Every value is verifiable against on-chain commits.

Mockup represents target operational state. Real numbers visible at vali-04.ddx.network once testnet is live.



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DEMO · GROUNDING IN ACTION

Prompt → bbox, scored live.

提示词 → 边界框，实时打分。

Four held-out prompts. The top-ranked miner's prediction overlaid on the source frame. Latency, IoU, and confidence settle to the right.

We are not inventing the mechanism. We are porting it.

我们没有发明机制，我们在移植它。

The architecture is ported from the **2025 hackathon Defektr proposal**. The commercial pattern is proven by **Score Vision (SN44)**, a deployed vision subnet with real enterprise ARR.

架构源自 2025 黑客松 Defektr 提案。商业模式由已部署的 Score Vision (SN44) 验证。

DEPLOYED PRECEDENT · SCORE VISION (SN44)

Real subnet. Real ARR. Same submission-and-scoring pattern.

真实部署。真实营收。同样的提交与评分模式。

Score Vision is a deployed Bittensor vision subnet running a submission-and-scoring marketplace. Different domain (sports video analysis), same architectural pattern as DDX. Proof that decentralized vision marketplaces convert technical progress into enterprise revenue.

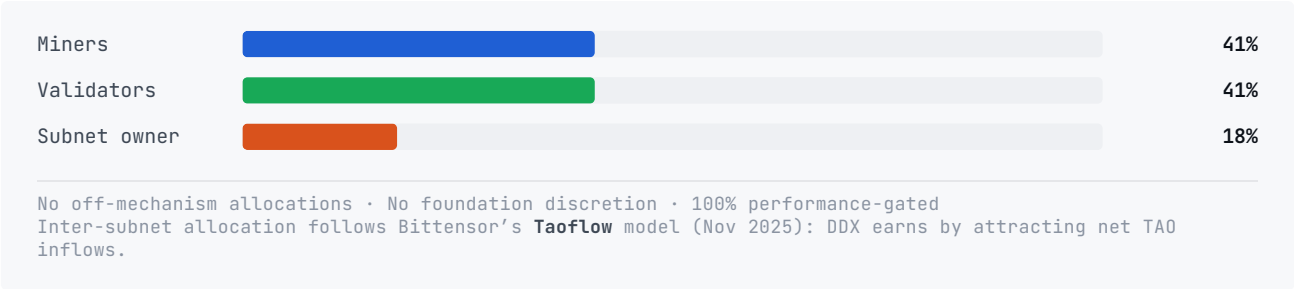
LIVE SINCE 2024	ENTERPRISE ARR ~\$3M
ENTERPRISE CLIENTS 7	AVG CONTRACT \$428K

EMISSIONS ALLOCATION · DDX

Where TAO goes.

TAO 奖励去哪里。

Standard 18 / 41 / 41 split between validators, miners, and subnet owner. Miner share routes 100% to held-out accuracy. No off-mechanism payouts, no “contribution” bonuses, no foundation discretion.



12-WEEK ROADMAP

Hackathon → testnet → mainnet → ship.

黑客松 → 测试网 → 主网 → 上线。

NOWWEEK 0 · HACKATHON

May 23 – May 26

Submission ready.

提交就绪。

- Validator binary v0.3 with ScreenSpot scoring loop
- Live dashboard at vali-04.ddx.network
- Baseline miner reproducing UI-TARS-7B
- This deck + 4-min stage demo

PHASE 1TESTNET

Week 1 – 4

Stand up testnet.

建立测试网。

- Validate scoring loop against UI-Vision, OSWorld-G held-out batches
- Add robustness multiplier + distribution-shift batch to validator
- Recruit 8 testnet validators (Score Vision ops, friends)
- Onboard 25 testnet miners with starter weights
- Adversarial review of anti-gaming surface

PHASE 2MAINNET

Week 5 – 8

Register sn-32. Open emissions.

注册 sn-32。开启奖励。

- Submit subnet registration. Accept the next available slot.
- Migrate held-out pool from testnet
- Stake-weighted Yuma trimmed-mean consensus live
- Public leaderboard launch

PHASE 3ECOSYSTEM

Week 9 – 12

Ship the API.

发布 API。

- HTTP grounding API gated to top-5 miners
- First 3 paying users (agent frameworks)
- Adopt Score Vision (SN44) integration playbook for enterprise sales
- v1.1 weight vote: VisualWebArena, DDX-private
- 50-validator decentralization milestone

• TEAM & ASK

Two operators. One mechanism. Shipping in 48 hours.

两个执行者。一套机制。48 小时内上线。

MECHANISM · VALIDATOR
机制 · 验证器

Drew Mimi

Owens the validator binary, the scoring function, and the anti-gaming surface. Writing every line of consensus-critical code with adversarial review built in.

Demo on stage · validator console
drew@ddx.network

MODELS · MINER STACK
模型 · 矿工栈

Keith Guo

Owens the baseline miner, the training data pipeline, and the open-weights release cadence. Shipping a competitive grounding miner benchmarked against **UI-TARS-72B** and **UI-Venus-72B** on day one.

Demo on stage · grounding model
keith@ddx.network

ASK · WHAT WE NEED FROM THIS HACKATHON

A subnet slot, validators, and miners.

一个子网名额、验证器和矿工。

- Bittensor Foundation — subnet registration green-light (any open slot)
- Score Vision ops (or another deployed vision subnet operator) — adversarial review of scoring loop
- 4 launch validators with >500 TAO stake
- Intros to UI-TARS, OS-Atlas, UI-Venus teams as launch miners
- \$50k to fund 8 weeks of validator infra + adversarial audit